

Experiment No.4

Study of GSM Modem and AT Commands

AIM: To study GSM Modem and AT Commands.

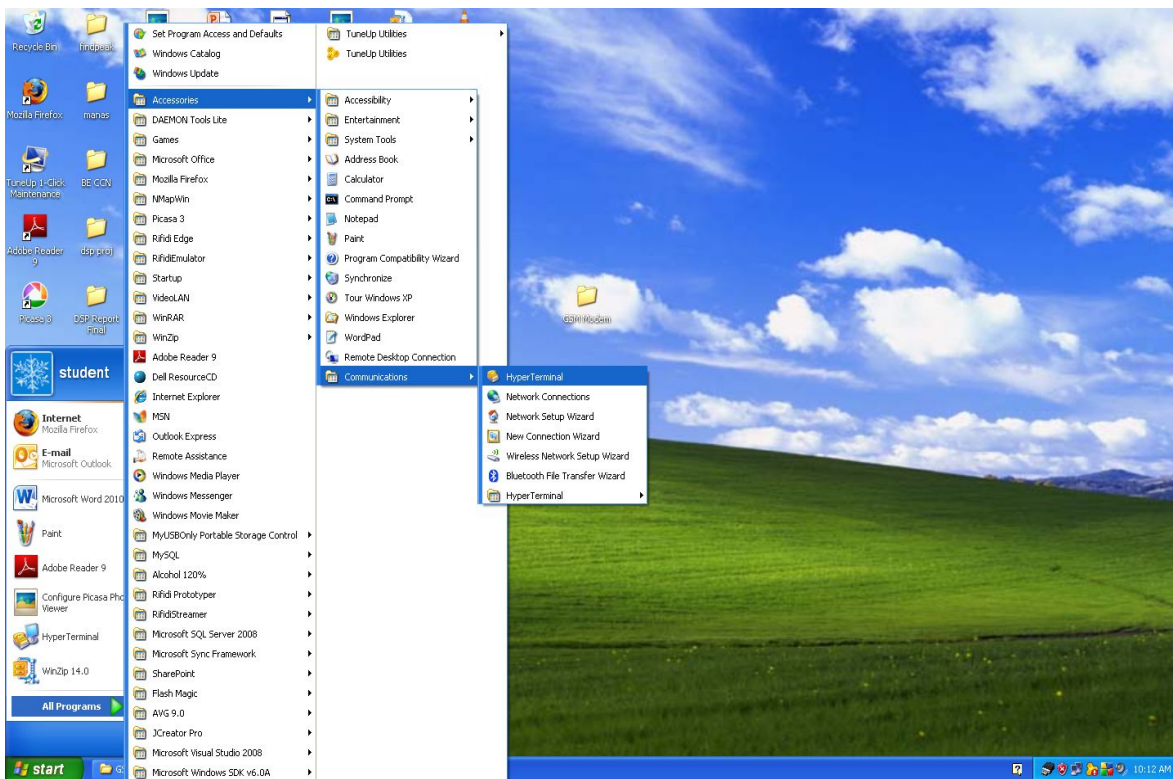
System Requirements: Hardware : GSM modem, PC, SIM card.
Software : Hyper Terminal.

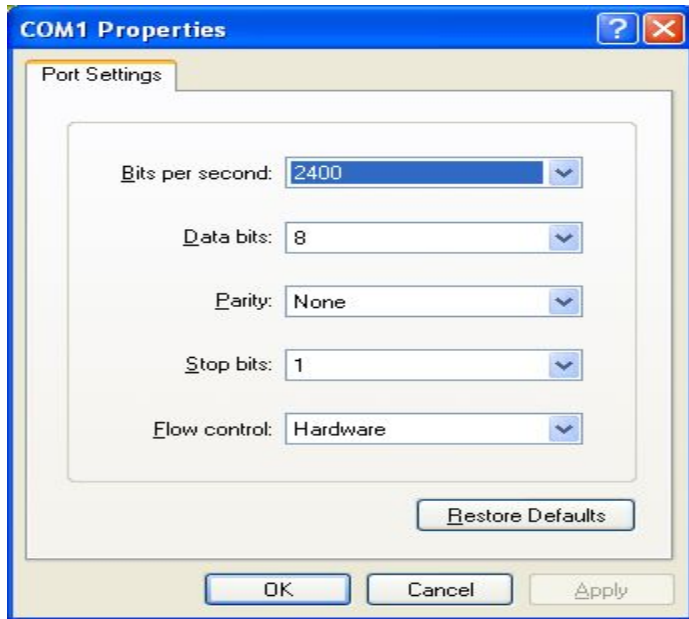
Introduction:

The NOKIA 30 is a GSM Connectivity Terminal with many advanced functions. With the Nokia 30 GSM Connectivity Terminal connected to your compatible PC or to a compatible device, you can, for example, make data calls, send messages, connected to the internet and so on.

The Nokia 30 offers high speed wireless connections and it supports several data connection types such as GPRS, high-speed data, USSD and text messages over EGSM900/GSM1800 networks.

When the Nokia 30 is attached to the data adapter RS-232, it can be used as a stand-alone modem when connected to a compatible PC or compatible devices.



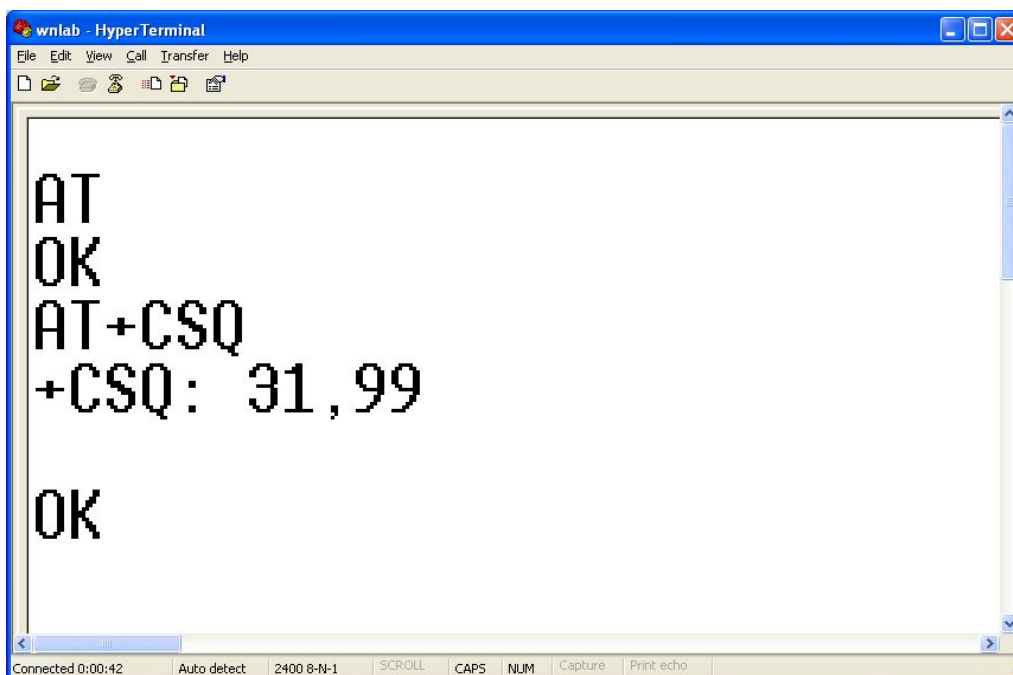


AT Commands :

1. Connect the modem to serial port and use hyper terminal to issue the commands to GSM modem

2. AT+CSQ

This command gives the signal strength of the GSM Network. Out put will be some number eg. 11,99 which indicates the signal strength network strength above 14 is considered as good at+csq enter



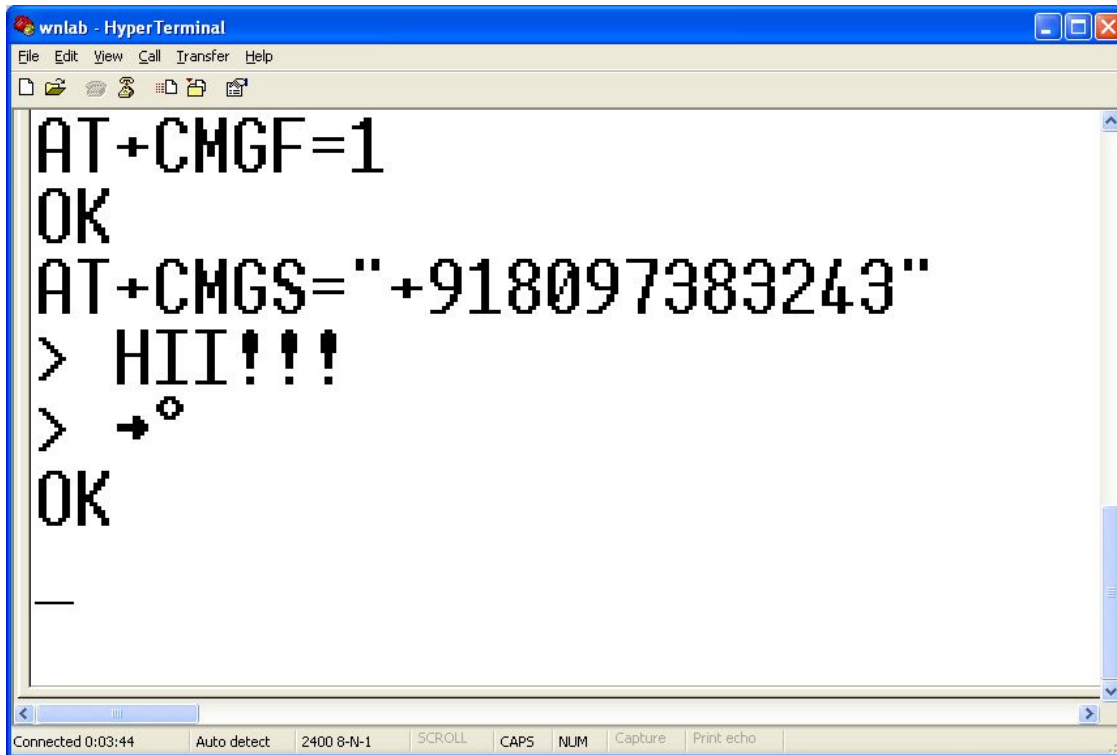
3. **AT+CMGF=<mode>**

For setting SMS mode i.e. PDU or TEXT

PDU = 0

TEXT = 1

at+cmgf=1 enter



The screenshot shows a HyperTerminal window titled 'wnlab - HyperTerminal'. The command history is as follows:

```
AT+CMGF=1
OK
AT+CMGS="+918097383243"
> HII!!!
> +°
OK
_
```

The status bar at the bottom indicates 'Connected 0:03:44', 'Auto detect', '2400 8-N-1', and various control buttons like SCROLL, CAPS, NUM, Capture, and Print echo.

4. **AT+CMGL=<stat> or AT+CMGR=<index>**

For reading SMS from the GSM terminal

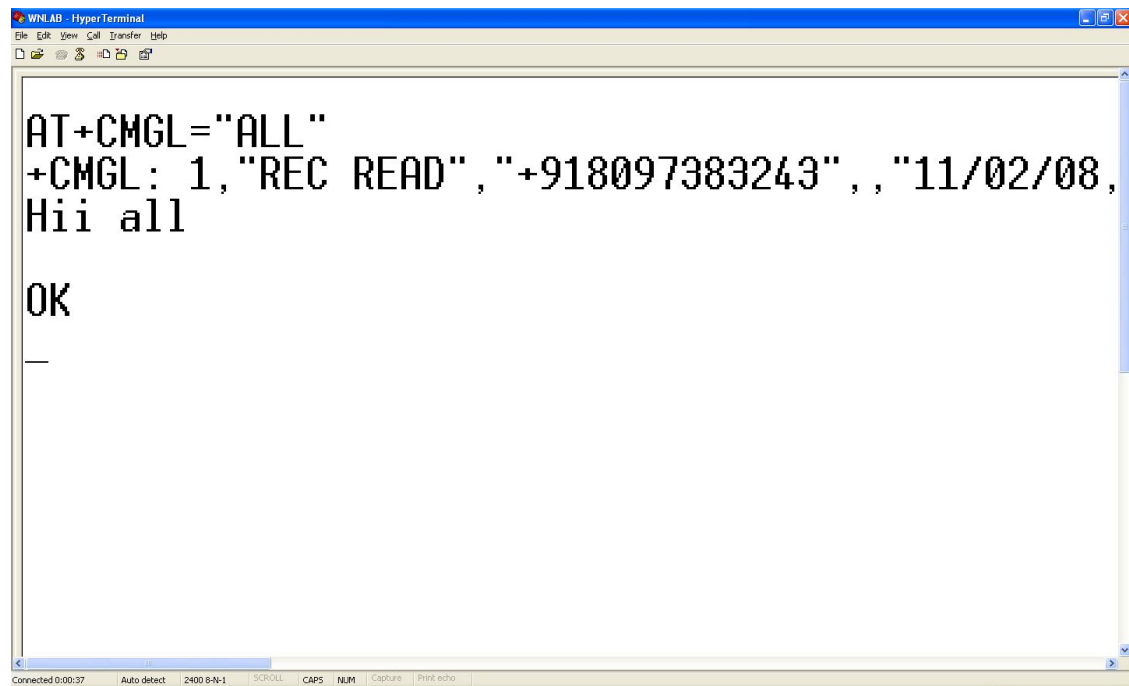
stat means status of the message

i.e. rec unread

rec read

at+cmgl="rec unread" this command will open msgs from the sim which are not read

at+cmgl="rec read" this command will open all the msgs from the sim which are read

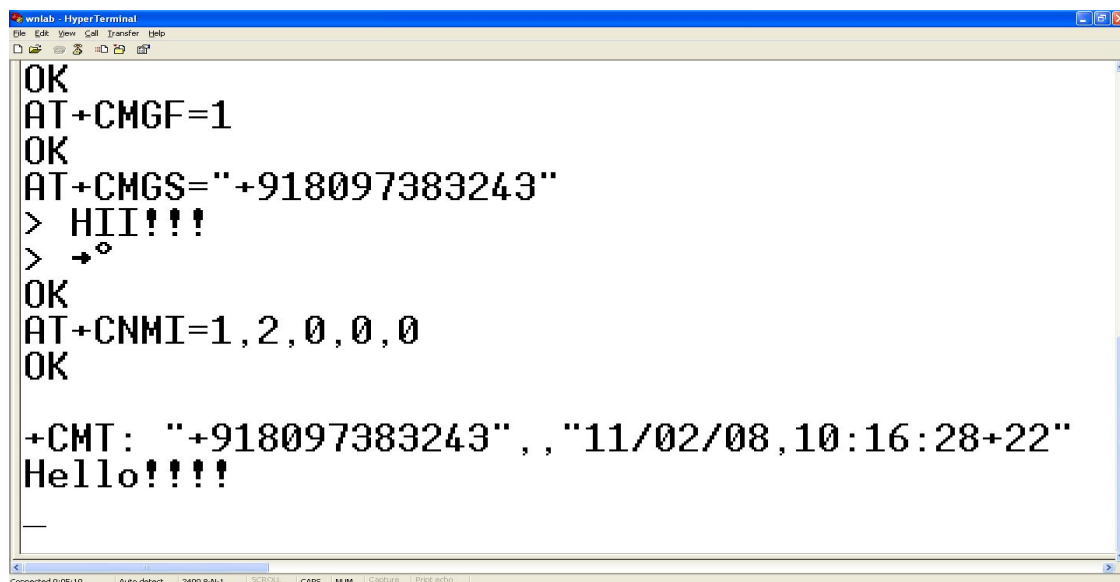


```
AT+CMGL="ALL"  
+CMGL: 1,"REC READ","+918097383243",,"11/02/08,  
Hii all  
  
OK
```

5. **AT+CMGS=<da><toda><cr>** text is entered here <ESC/ctrl-z>

For sending Messages
da : destination number
toda: 145

```
at+cmgs="+919819134064",145<cr>  
> type your msg here  < to send msg press ctrl+z >
```



```
OK  
AT+CMGF=1  
OK  
AT+CMGS="+918097383243"  
> HII!!!  
> ↵  
OK  
AT+CNMI=1,2,0,0,0  
OK  
  
+CMT: "+918097383243",,"11/02/08,10:16:28+22"  
Hello!!!!
```

6. **AT+CMGD=<index>**

For deleting message
index : msg no
at+cmgd=14
this command will delete msg no. 14 from the sim

CONCLUSION: